A MAP OF PRESHEAVES WHICH IS BIJECTIVE ON A COVERING FAMILY BECOMES AN ISOMORPHISM AFTER SHEAFIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Sections of a sheafification. Same chapter, Lemma lemma-sections-sheafification. Describes the sections of a sheafification by coverings and local sections, which is the tool used in both directions.

Source reference. Original lemma 00WK: sources/stacks-sites/pinned-original-source.tex, source lines 2410–2410. Navigation pointer invoked at proof line 2410; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{F} \to \mathcal{G}$ be a map of presheaves of sets on $\mathcal{C}$. Denote $\mathcal{B}$ the set of $U \in \mathrm{Ob}(\mathcal{C})$ such that $\mathcal{F}(U) \to \mathcal{G}(U)$ is bijective. If every object of $\mathcal{C}$ has a covering by elements of $\mathcal{B}$, then $\mathcal{F}^\# \to \mathcal{G}^\#$ is an isomorphism.*

Proof. Let $U \in \mathrm{Ob}(\mathcal{C})$. Let us prove that $\mathcal{F}^\#(U) \to \mathcal{G}^\#(U)$ is surjective. To do this we will use Lemma 00WK without further mention. For any $s \in \mathcal{G}^\#(U)$ there exists a covering $\{U_i \to U\}$ and sections $s_i \in \mathcal{G}(U_i)$ such that

(1) $s|_{U_i}$ is the image of s_i via $\mathcal{G} \to \mathcal{G}^\#$, and
(2) for every i, j there exists a covering $\{U_{ijk} \to U_i \times_U U_j\}$ of $\mathcal{D}$ such that the pullbacks of s_i and s_j to each U_{ijk} agree.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

By assumption, for each i we may choose a covering $\{U_{i,a} \to U_i\}$ with $U_{i,a} \in \mathcal{B}$. Then $\{U_{i,a} \to U\}$ is a covering. Denoting $s_{i,a}$ the image of s_i in $\mathcal{G}(U_{i,a})$ we see that the pullbacks of $s_{i,a}$ and $s_{j,b}$ to the members of the covering

$$\{(U_{i,a} \times_U U_{j,b}) \times_{U_i \times_U U_j} U_{ijk} \to U_{i,a} \times_U U_{j,b}\}$$

agree. Hence we may assume that $U_i \in \mathcal{B}$. Repeating the argument, we may also assume $U_{ijk} \in \mathcal{B}$ for all i, j, k (details omitted). Then since $\mathcal{F}(U_i) \to \mathcal{G}(U_i)$ is bijective by our definition of $\mathcal{B}$ in the statement of the lemma, we get unique $t_i \in \mathcal{F}(U_i)$ mapping to s_i. The pullbacks of t_i and t_j to each U_{ijk} agree in $\mathcal{F}(U_{ijk})$ because $U_{ijk} \in \mathcal{B}$ and because we have the agreement for t_i and t_j. Then the lemma tells us there exists a unique $t \in \mathcal{F}^\#(U)$ such that $t|_{U_i}$ is the image of t_i. It follows that t maps to s. We omit the proof that $\mathcal{F}^\#(U) \to \mathcal{G}^\#(U)$ is injective. $\square$